21 (a) Write 152 as a product of its prime factors.

$152 \div 2 = 76$

$76 \div 2 = 38$

$38 \div 2 = 19$ (prime)

$152 = 2 \times 2 \times 2 \times 19 = 2^3 \times 19$

2 marks total
Divide repeatedly by primes; write a product.

$2^3 \times 19$

(2)

(b) Find the highest common factor (HCF) of 96 and 152

$96 = 2 \times 2 \times 2 \times 2 \times 2 \times 3$

$152 = 2 \times 2 \times 2 \times 19$

Common prime factors: $2 \times 2 \times 2 = 8$

2 marks total
Multiply the shared prime factors.

8

(2)

(Total for Question 21 is 4 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA